Daily Work Report – DevOps Intern

Week 1

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Date:

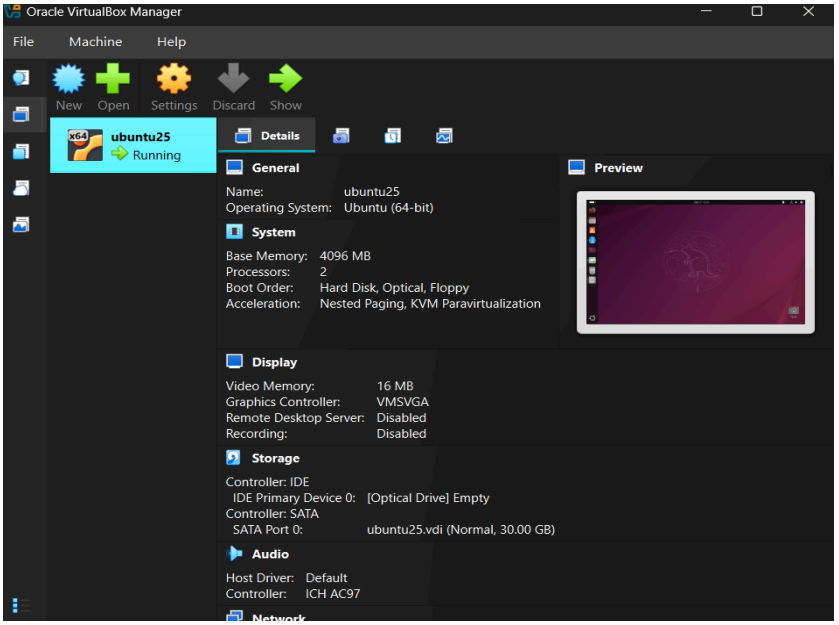
Day 0 - 12/12/2025

Tasks Completed Today:

- **Task1:** Installed ubuntu VM using Virtual box on windows OS.
- **Task2:** Learned basic linux commands like pwd, ls, cat, tail, head, touch, mkdir, cp, mv, wc, ln, locate, echo etc. and executed them in the terminal.
- **Task3:** Learned to use vim and nano text editors in linux. Learned to add and modify text using these editors in the terminal.

Learnings:

- Steps to install ubuntu VM using Virtual box on windows OS.
    - Download the latest ubuntu ISO image from browser
    - Launch Oracle VM VirtualBox Manage from the Start Menu.
    - Click New. Define name and the downloaded ISO image.Specify OS Version.
    - Allocate System Memory (RAM), CPU core, Disk size.
    - Click Finish setup.



- Basic linux commands:
  - pwd: Displays the full path of the current directory you are in.

```
ubuntuuser@ubuntu25:~$ pwd
/home/ubuntuuser
ubuntuuser@ubuntu25:~$
```

  - ls: Lists all files and directories in the current location.

```
ubuntuuser@ubuntu25:/etc$ ls
ModemManager          cron.d                gdb                   kdump                 network               rc5.d                 supercat
NetworkManager        cron.daily            gdm3                  kernel                networkd-dispatcher   rc6.d                 sysctl.d
PackageKit            cron.hourly           geoclue               krbs.conf.d           networks              rcS.d                 sysstat
UPower                cron.monthly          ghostscript           ld.so.cache           newt                  resolv.conf           systemd
X11                   cron.weekly           glvnd                 ld.so.conf            nftables.conf         rmt                   terminfo
adduser.conf          cron.yearly           gnome                 ld.so.conf.d          nsswitch.conf         rpc                   thermald
alsa                  crontab               gnome-remote-desktop  ldap                  openvpn               rsyslog.conf          timezone
alternatives          cups                  gnutls                legal                 opt                   rsyslog.d             tmpfiles.d
anacrontab            cupshelpers           gprofng.rc            libao.conf            os-release            rygel.conf            ubuntu-advantage
apm                   dbus-1                groff                 libaudit.conf         pam.conf              sane.d                ucf.conf
apparmor              dconf                 group                 libblockdev           pam.d                 screenrc              udev
apparmor.d            debconf.conf          group-                libibverbs.d          paperspecs            security              udisks2
apport                debian_version        grub.d                libnl-3               passwd                selinux               ufw
apt                   debuginfod            gshadow               locale.conf           passwd-               sensors.d             update-manager
avahi                 default               gshadow-              locale.gen            pcmcia                sensors3.conf         update-motd.d
bash.bashrc           deluser.conf          gss                   localtime             perl                  services              update-notifier
bash_completion       depmod.d              gtk-2.0               logcheck              pki                   sgml                  usb_modeswitch.conf
bash_completion.d     dhcp                  gtk-3.0               login.defs            plymouth              shadow                usb_modeswitch.d
bindresvport.blacklist dhcpcd.conf          hdparm.conf           logrotate.conf        pm                    shadow-               vconsole.conf
binfmt.d              dictionaries-common   host.conf             logrotate.d           pnm2ppa.conf          shells                vim
bluetooth             dpkg                  hostname              lsb-release           polkit-1              skel                  vtrgb
brlapi.key            dracut.conf           hosts                 machine-id            ppp                   snmp                  vulkan
brltty                dracut.conf.d         hosts.allow           magic                 printcap              speech-dispatcher     wgetrc
brltty.conf           e2scrub.conf          hosts.deny            magic.mime            profile               ssh                   wpa_supplicant
ca-certificates       emacs                 hp                    manpath.config        profile.d             ssl                   xattr.conf
ca-certificates.conf  environment           ifplugd               mime.types            protocols             sssd                  xdg
chatscripts           environment.d         init                  mke2fs.conf           pulse                 subgid                xml
chrony                ethertypes            init.d                modprobe.d            python3               subgid-               zsh_command_not_found
cloud                 fonts                 initramfs-tools       modules               python3.13            subuid
colord                fprintd.conf          inputrc               modules-load.d        rc0.d                 subuid-
console-setup         fstab                 inserv.conf.d         mtab                  rc1.d                 sudo.conf
cracklib              fuse.conf             ipp-usb               nanorc                rc2.d                 sudo_logsrvd.conf
credstore             fwupd                 issue                 netconfig             rc3.d                 sudoers
credstore.encrypted   gai.conf              issue.net             netplan               rc4.d                 sudoers.d
```

    - ls -l: Shows in list format

```
ubuntuuser@ubuntu25:~$ ls -l Documents/csttasks/
total 8
-rw-rw-r-- 1 ubuntuuser ubuntuuser   0 Dec 12 08:45 KAILAS
-rw-rw-r-- 1 ubuntuuser ubuntuuser   0 Dec 12 08:44 TASK
-rw-rw-r-- 1 ubuntuuser ubuntuuser   0 Dec 12 07:59 file1
-rw-rw-r-- 1 ubuntuuser ubuntuuser  67 Dec 12 10:14 file2
-rw-rw-r-- 1 ubuntuuser ubuntuuser   0 Dec 12 08:00 file3
-rw-rw-r-- 1 ubuntuuser ubuntuuser   0 Dec 12 08:00 file4
-rw-rw-r-- 1 ubuntuuser ubuntuuser   0 Dec 12 08:00 file5
-rw-rw-r-- 1 ubuntuuser ubuntuuser   0 Dec 12 08:00 file6
-rw-rw-r-- 1 ubuntuuser ubuntuuser   0 Dec 12 08:45 kailas
-rw-rw-r-- 2 ubuntuuser ubuntuuser 133 Dec 12 10:18 sample.txt
-rw-rw-r-- 1 ubuntuuser ubuntuuser   0 Dec 12 08:44 task
```

    - ls -a: Show hidden files
    - ls -R: Shows recursive listing i.e., directories along with subdirectories.

```
ubuntuuser@ubuntu25:~$ ls -R Documents
Documents:
csttasks

Documents/csttasks:
KAILAS  TASK  file1  file2  file3  file4  file5  file6  kailas  sample.txt  task
ubuntuuser@ubuntu25:~$
```

    - ls -la: Shows hidden files with list format.
    - ls -lr: Shows files in reverse order.

```
ubuntuuser@ubuntu25:~$ ls -R Documents
Documents:
csttasks

Documents/csttasks:
KAILAS  TASK  file1  file2  file3  file4  file5  file6  kailas  sample.txt  task
ubuntuuser@ubuntu25:~$
```

  - cat: Used to view the content of a file in the terminal.

```
ubuntuuser@ubuntu25:/etc$ cat adduser.conf
# /etc/adduser.conf: `adduser' configuration.
# See adduser(8) and adduser.conf(5) for full documentation.

# A commented out setting indicates that this is the default in the
# code. If you need to change those settings, remove the comment and
# make your intended change.

# STDERRMSGLEVEL, STDOUTMSGLEVEL, and LOGMSGLEVEL set the minimum
# priority for messages logged to syslog/journal and the console,
# respectively.
# Values are trace, debug, info, warn, err, and fatal.
# Messages with the set priority or higher get logged to the
# respective medium.
#STDOUTMSGLEVEL=warn
#STDERRMSGLEVEL=warn
#LOGMSGLEVEL=info

# The login shell to be used for all new users.
# Default: DSHELL=/bin/bash
#DSHELL=/bin/bash

# The directory in which new home directories should  be  created.
# Default: DHOME=/home
# DHOME=/home

# The directory from which skeletal user configuration files
# will be copied.
# Default: SKEL=/etc/skel
#SKEL=/etc/skel

# Specify inclusive ranges of UIDs and GIDs from which UIDs and GIDs
# for system users, system groups, non-system users and non-system groups
# can be dynamically allocated.
# Default: FIRST_SYSTEM_UID=100, LAST_SYSTEM_UID=999
```

- tail : View the End of a File. Shows the last 10 lines of a file by default. We can specify the number of lines to be seen by using -n flag.
- head: View the Beginning of a File. Shows the first 10 lines of a file.

```
ubuntuuser@ubuntu25:/etc$ tail -n3 adduser.conf

# use extrausers by default
#USE_EXTRAUSERS=1
ubuntuuser@ubuntu25:/etc$ head -n5 adduser.conf
# /etc/adduser.conf: `adduser' configuration.
# See adduser(8) and adduser.conf(5) for full documentation.

# A commented out setting indicates that this is the default in the
# code. If you need to change those settings, remove the comment and
```

- touch: Creates an empty file. Can create more than one file with a single touch command and specifying the filenames with a space.
  - rm: To remove the file.
  - rm -f: forcefully delete.
  - rm -i: Asks for confirmation before deletion.
  - rm -r: Recursive delete. Deletes the directory with its subfolders.
  - rm -rf: Recursive and force delete.

```
ubuntuuser@ubuntu25:~$ cd Documents/csttasks/
ubuntuuser@ubuntu25:~/Documents/csttasks$ touch file1
ubuntuuser@ubuntu25:~/Documents/csttasks$ touch file2 file3 file4 file5 file6
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls
file1  file2  file3  file4  file5  file6
ubuntuuser@ubuntu25:~/Documents/csttasks$
```

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ rm -i KAILAS
rm: remove regular empty file 'KAILAS'? Y
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l
total 12
-rw-rw-r-- 1 ubuntuuser ubuntuuser    0 Dec 12 08:44 TASK
-rw-rw-r-- 1 ubuntuuser ubuntuuser    0 Dec 12 07:59 file1
-rw-rw-r-- 1 ubuntuuser ubuntuuser   67 Dec 12 10:14 file2
-rw-rw-r-- 1 ubuntuuser ubuntuuser    0 Dec 12 08:00 file3
-rw-rw-r-- 1 ubuntuuser ubuntuuser    0 Dec 12 08:00 file4
-rw-rw-r-- 1 ubuntuuser ubuntuuser    0 Dec 12 08:00 file5
-rw-rw-r-- 1 ubuntuuser ubuntuuser    0 Dec 12 08:00 file6
drwxrwxr-x 3 ubuntuuser ubuntuuser 4096 Dec 12 11:48 file7
-rw-rw-r-- 1 ubuntuuser ubuntuuser    0 Dec 12 08:45 kailas
-rw-rw-r-- 2 ubuntuuser ubuntuuser  133 Dec 12 10:18 sample.txt
-rw-rw-r-- 1 ubuntuuser ubuntuuser    0 Dec 12 08:44 task
ubuntuuser@ubuntu25:~/Documents/csttasks$ rm -v kailas
removed 'kailas'
ubuntuuser@ubuntu25:~/Documents/csttasks$ rm -f file4
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l
total 12
-rw-rw-r-- 1 ubuntuuser ubuntuuser    0 Dec 12 08:44 TASK
-rw-rw-r-- 1 ubuntuuser ubuntuuser    0 Dec 12 07:59 file1
-rw-rw-r-- 1 ubuntuuser ubuntuuser   67 Dec 12 10:14 file2
-rw-rw-r-- 1 ubuntuuser ubuntuuser    0 Dec 12 08:00 file3
-rw-rw-r-- 1 ubuntuuser ubuntuuser    0 Dec 12 08:00 file5
-rw-rw-r-- 1 ubuntuuser ubuntuuser    0 Dec 12 08:00 file6
drwxrwxr-x 3 ubuntuuser ubuntuuser 4096 Dec 12 11:48 file7
-rw-rw-r-- 2 ubuntuuser ubuntuuser  133 Dec 12 10:18 sample.txt
-rw-rw-r-- 1 ubuntuuser ubuntuuser    0 Dec 12 08:44 task
```

- mkdir: Make directory command. Creates one or more directories or folders.
  - rmdir: Used to delete the directory
  - mkdir -p: Makes a new directory with its subfolders if wanted.

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ mkdir file7
ubuntuuser@ubuntu25:~/Documents/csttasks$ mkdir -p file7/8/9/10
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -R
.:
KAILAS  TASK  file1  file2  file3  file4  file5  file6  file7  kailas  sample.txt  task

./file7:
8

./file7/8:
9

./file7/8/9:
10

./file7/8/9/10:
```

- cp: To copy files from one location to another.
  - cp source destination

```
ubuntuuser@ubuntu25:~$ cp Documents/csttasks/file1 Downloads
ubuntuuser@ubuntu25:~$ ls Downloads/
file1  link1
```

- mv : To move files from one location to another.
  - Eg: mv sourcefile destinationfile. If the destinationfile doesn't exist , mv commands renames the sourcefile to the destinationfile.

```
ubuntuuser@ubuntu25:~$ mv Documents/csttasks/FILE1 Downloads/
ubuntuuser@ubuntu25:~$ ls Documents/csttasks/
KAILAS  TASK  file1  file2  file3  file4  file5  file6  kailas  sample.txt  task
ubuntuuser@ubuntu25:~$ ls Downloads/
FILE1  file1  link1
ubuntuuser@ubuntu25:~$
```

```
ubuntuuser@ubuntu25:~$ ls Downloads/
FILE1  file1  link1
ubuntuuser@ubuntu25:~$ cd Downloads
ubuntuuser@ubuntu25:~/Downloads$ mv FILE1 FILE2
ubuntuuser@ubuntu25:~/Downloads$ ls
FILE2  file1  link1
ubuntuuser@ubuntu25:~/Downloads$
```

- wc: Word Count. This command counts the lines, words and characters in a file.
  - wc -l filename: to count lines.
  - wc -w filename: to count words.
  - wc -m filename: to count characters.
  - wc -c filename: to count bytes.
  - wc -L filename: to count the longest line. Gives the number of word in the longest line in a file.

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ wc /etc/adduser.conf
 118  528 4027 /etc/adduser.conf
ubuntuuser@ubuntu25:~/Documents/csttasks$ wc -l /etc/adduser.conf
118 /etc/adduser.conf
ubuntuuser@ubuntu25:~/Documents/csttasks$ wc -w /etc/adduser.conf
528 /etc/adduser.conf
ubuntuuser@ubuntu25:~/Documents/csttasks$ wc -c /etc/adduser.conf
4027 /etc/adduser.conf
ubuntuuser@ubuntu25:~/Documents/csttasks$ wc -m /etc/adduser.conf
4027 /etc/adduser.conf
ubuntuuser@ubuntu25:~/Documents/csttasks$ wc -L /etc/adduser.conf
75 /etc/adduser.conf
```

- In: Creates links. Links can be hard and soft.
  - Hard link: Used as a backup. Creates a backup file of the source file. Whatever content is modified in the source will also be modified in the backup file. Even if the source is lost the backup remains.

```
ubuntuuser@ubuntu25: ~

ubuntuuser@ubuntu25:~/Documents/csttasks$ cat sample.txt
This is a sample text
Whatever modifications made will be backed up
ubuntuuser@ubuntu25:~/Documents/csttasks$ cd
ubuntuuser@ubuntu25:~$ ln Documents/csttasks/sample.txt backup/backup.txt
ubuntuuser@ubuntu25:~$ cat backup/backup.txt
This is a sample text
Whatever modifications made will be backed up
ubuntuuser@ubuntu25:~$
```

```
ubuntuuser@ubuntu25:~$ echo "This is the new modification" >> Documents/csttasks/sample.txt
ubuntuuser@ubuntu25:~$ cat backup/backup.txt
This is a sample text
Whatever modifications made will be backed up
This is the new modification
ubuntuuser@ubuntu25:~$
```

  - Soft link: It is a symbolic link. More like a shortcut. It doesn't show the content but points to the sourcefile.

```
ubuntuuser@ubuntu25:~$ ln -s Documents/csttasks/TASK Downloads/link1
ubuntuuser@ubuntu25:~$ ls -l Downloads/
total 0
lrwxrwxrwx 1 ubuntuuser ubuntuuser 23 Dec 12 09:44 link1 -> Documents/csttasks/TASK
ubuntuuser@ubuntu25:~$
```

- locate: Used to search for files. Finds files and directories by name using a pre-built database.

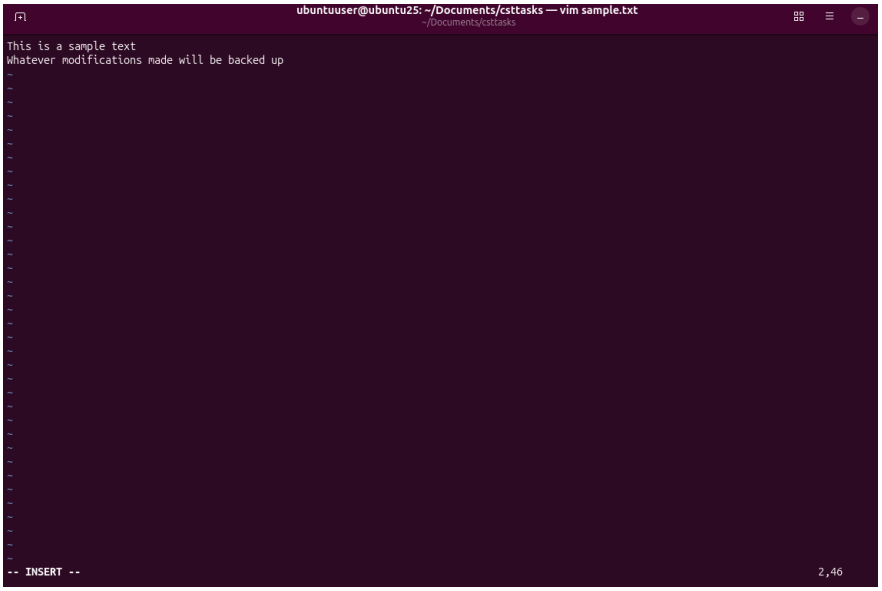
```
ubuntuuser@ubuntu25:~$ locate file1
/home/ubuntuuser/Documents/csttasks/file1
/usr/share/doc/libsndfile1
/usr/share/doc/libxkbfile1
/usr/share/doc/libsndfile1/changelog.Debian.gz
/usr/share/doc/libsndfile1/copyright
/usr/share/doc/libxkbfile1/changelog.Debian.gz
/usr/share/doc/libxkbfile1/copyright
/var/lib/dpkg/info/libsndfile1:amd64.list
/var/lib/dpkg/info/libsndfile1:amd64.md5sums
/var/lib/dpkg/info/libsndfile1:amd64.shlibs
/var/lib/dpkg/info/libsndfile1:amd64.symbols
/var/lib/dpkg/info/libsndfile1:amd64.triggers
/var/lib/dpkg/info/libxkbfile1:amd64.list
/var/lib/dpkg/info/libxkbfile1:amd64.md5sums
/var/lib/dpkg/info/libxkbfile1:amd64.shlibs
/var/lib/dpkg/info/libxkbfile1:amd64.triggers
ubuntuuser@ubuntu25:~$ locate -c file1
16
ubuntuuser@ubuntu25:~$
```

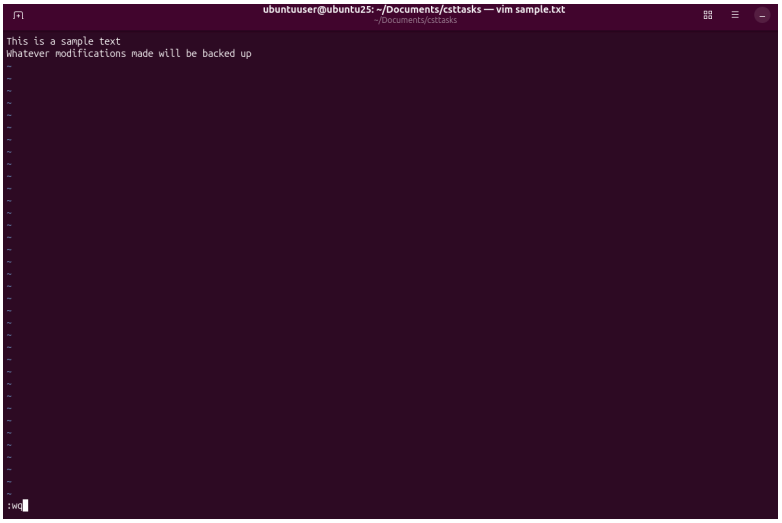
  - locate -c filename: Shows only number of matches.

        - locate -i filename: Does case- insensitive search.
    - echo : Used to print. Displays a message or the value of a variable.
        - echo - n: to avoid newline
        - echo -e: Enable escape characters**.** Allows special characters like \n, \t
        - To print environment variables we use $. For instance: echo $path with print the path as the output.

```
ubuntuuser@ubuntu25:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/snap/bin
ubuntuuser@ubuntu25:~$ echo -n "hello"
helloubuntuuser@ubuntu25:~$ echo -e "hello\nworld"
hello
world
ubuntuuser@ubuntu25:~$ echo -e "hello\tworld"
hello	world
ubuntuuser@ubuntu25:~$ echo "Kailas"
Kailas
ubuntuuser@ubuntu25:~$
```

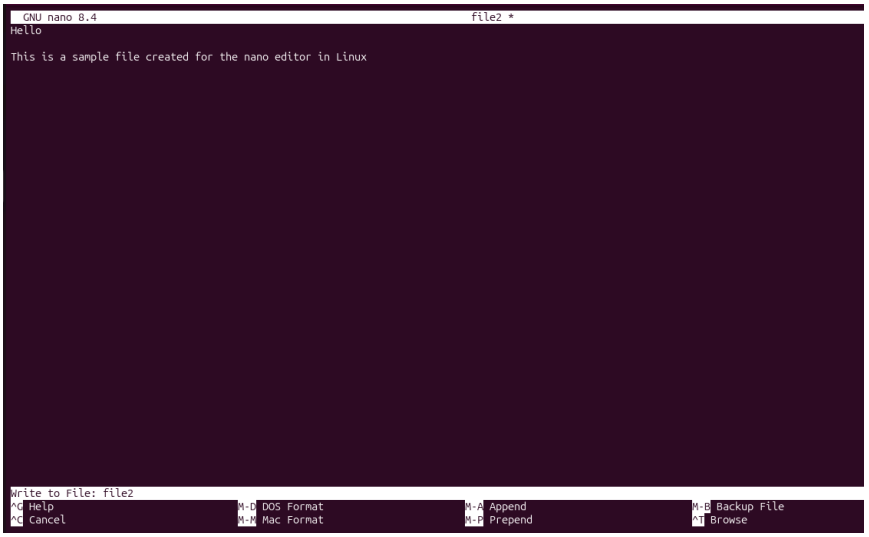
- Vim editor:
    - Opening vim: vim filename
    - Modes in Vim
    - Vim works in 3 main modes:
        - Normal Mode (default)
            - Used for navigation & commands
            - Press Esc anytime to return here
        - Insert Mode
            - Used for typing text. Press i.
        - Command Mode
            - For saving, quitting, searching
        - Basic Editing
            - x: Delete a character
            - dd: Delete a line
            - yy: Copy (yank) a line
            - u: Undo
            - Ctrl + r: Redo
            - 0: Go to start of line
            - $: Go to end of line
            - gg: Go to top
            - G: Go to bottom
            - :5: Go to line number 5
            - :w: Save
            - :q: Quit
            - :wq: Save and quit
            - :q!: Quit without saving





- Nano editor:
  - Opening Nano: nano filename
  - Basic Editing:
    - No modes like vim
    - Ctrl + O: Save (Write Out)
    - Ctrl + X: Exit
    - Ctrl + O, Enter: Save and continue
    - Ctrl + K: Cut line
    - Ctrl + U: Paste
    - Alt + 6: Copy line
    - Alt + U: Undo
    - Alt + E: Redo

- Ctrl + W: Search
- Ctrl + W, then Ctrl + R: Replace text
- Ctrl + A: Move to start of line
- Ctrl + E: Move to end of line
- Ctrl + Y: Page up
- Ctrl + V: Page down



Day 1 - 15/12/2025

Tasks Completed Today:

- **Task1:** Familiarized network commands netstat, ss, ifconfig, traceroute and its uses. Executed and tested it in the terminal.
- **Task2:** Familiarized monitoring commands ps, top, htop, iotop. Tested its real time customization on the terminal.
- **Task3:** Familiarized text manipulation commands: awk, sed. Also familiarized grep,tail,head and executed them in terminal.

Learnings:

- **Network Commands:**
    - **netsat:** Used to display network connections, routing tables, interface statistics, and listening ports.

```
ubuntuuser@ubuntu25:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 ubuntu25:bootpc         _gateway:bootps         ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     19798    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     21128    /run/dbus/system_bus_socket
unix  3      [ ]         DGRAM      CONNECTED     1838
unix  3      [ ]         STREAM     CONNECTED     18373
unix  2      [ ]         DGRAM                    11253
unix  3      [ ]         STREAM     CONNECTED     21771    /home/ubuntuuser/.cache/ibus/dbus-QTOubdqh
unix  2      [ ]         STREAM     CONNECTED     20220
unix  3      [ ]         STREAM     CONNECTED     21059
unix  3      [ ]         STREAM     CONNECTED     19973    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     12503    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     22149    /run/user/1000/at-spi/bus
unix  3      [ ]         STREAM     CONNECTED     20263    /run/user/1000/pipewire-0
unix  3      [ ]         STREAM     CONNECTED     21004
unix  3      [ ]         STREAM     CONNECTED     18118    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     18730    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     20333    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     20222
unix  3      [ ]         STREAM     CONNECTED     12495
unix  3      [ ]         DGRAM      CONNECTED     4510
unix  3      [ ]         STREAM     CONNECTED     21735    /run/user/1000/gvfsd/socket-RhV0HclQ
unix  3      [ ]         STREAM     CONNECTED     21437    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     21069    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     21058
unix  3      [ ]         STREAM     CONNECTED     18377    /run/user/1000/pipewire-0
unix  3      [ ]         STREAM     CONNECTED     18143
unix  3      [ ]         STREAM     CONNECTED     19504    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     22553    /run/snapd-snap.socket
unix  3      [ ]         STREAM     CONNECTED     22529
unix  3      [ ]         STREAM     CONNECTED     12629
unix  3      [ ]         STREAM     CONNECTED     28983    /run/user/1000/bus
```

        - Terms:
            - **Proto:** Defines the protocol used(TCP/UDP). Here UDP is the protocol.
            - **Send-Q, Recv-Q:** Indicates the data waiting to be sent and the data received but not yet read. In this case 0 indicates that there no pending data to be sent or to be read.

- **Local address:** Port and IP address of the local machine.
- **Foreign address:** Port and IP address of the remote system.
- **State:** Current connection state. **ESTABLISHED** indicated DHCP connection is active
- **Active UNIX domain sockets:**UNIX sockets are used for local communication inside the same system.

- **netstat -tuln:** Shows active listening ports.
  - -t: Shows TCP connections.
  - -u: Shows UDP connections.
  - -l: Shows listening ports. Listening services are clearly marked with **State = LISTEN** .
  - -n: Shows numeric IPs and ports. That is IP shown as numbers instead of names which is faster.

```
ubuntuuser@ubuntu25:~$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN
tcp6       0      0 ::1:631                 :::*                    LISTEN
udp        0      0 127.0.0.54:53           0.0.0.0:*
udp        0      0 127.0.0.53:53           0.0.0.0:*
udp        0      0 127.0.0.1:323           0.0.0.0:*
udp        0      0 0.0.0.0:45475           0.0.0.0:*
udp        0      0 0.0.0.0:60636           0.0.0.0:*
udp        0      0 0.0.0.0:5353            0.0.0.0:*
udp        0      0 0.0.0.0:52498           0.0.0.0:*
udp        0      0 10.0.2.15:3702          0.0.0.0:*
udp        0      0 239.255.255.250:3702    0.0.0.0:*
udp6       0      0 ::1:323                 :::*
udp6       0      0 :::55753                :::*
udp6       0      0 :::41774                :::*
udp6       0      0 :::5353                 :::*
udp6       0      0 fe80::a00:27ff:fe9:3702 :::*
udp6       0      0 ff02::c:3702            :::*
```

- netstat -r: Shows routing tables. Tells the system where to send packets based on the destination IP address.
- netstat -i: Shows network interfaces. Tells where the date enters or exits.

```
ubuntuuser@ubuntu25:~$ netstat  -r
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
default         _gateway        0.0.0.0         UG        0 0          0 enp0s3
10.0.2.0        0.0.0.0         255.255.255.0   U         0 0          0 enp0s3
ubuntuuser@ubuntu25:~$ netstat -i
Kernel Interface table
Iface             MTU    RX-OK RX-ERR RX-DRP RX-OVR    TX-OK TX-ERR TX-DRP TX-OVR Flg
enp0s3           1500    95223      0      0 0         39376      0      0      0 BMRU
lo              65536     3685      0      0 0          3685      0      0      0 LRU
```

- **enp0s3** is the main ethernet network interface. **lo** is loopback interface. Used for internal connections.

- **ss:** Used to display detailed information about sockets. Faster and more efficient than netstat. Commonly used to check listening ports, established connections, and TCP/UDP details.
  - **ss -tuln:** Shows listening ports as in netstat -tuln.
  - **ss -tn:** Shows established connections.
  - To check if a specific port is listening or not we can use grep with pipe command. Eg: ss -tuln | grep 80. This checks whether any service is listening on port 80.

```
ubuntuuser@ubuntu25:~$ ss
RTNETLINK answers: Invalid argument
Netid     State      Recv-Q     Send-Q                                          Local Address:Port                 Peer Address:Port
u_str     ESTAB      0          0                                                             * 51383                            * 51384
u_seq     ESTAB      0          0                                                             * 42111                            * 42110
u_str     ESTAB      0          0                                     /run/dbus/system_bus_socket 19798                          * 19797
u_seq     ESTAB      0          0                                                             * 41427                            * 41426
u_str     ESTAB      0          0                                                             * 40305                            * 40306
u_str     ESTAB      0          0                                     /run/dbus/system_bus_socket 21128                          * 20226
u_dgr     ESTAB      0          0                                                             * 1838                             * 1837
u_str     ESTAB      0          0                                                             * 18373                            * 18374
u_seq     ESTAB      0          0                                                             * 43567                            * 43566
u_str     ESTAB      0          0                          /home/ubuntuuser/.cache/ibus/dbus-QTOubdqh 21771                      * 21770
u_str     ESTAB      0          0                                                             * 20220                            * 0
u_str     ESTAB      0          0                                                             * 21059                            * 21068
u_str     ESTAB      0          0                                                 /run/user/1000/bus 19873                       * 19872
```

  - **Terms:**
    - **Netid:** Socket type. For example, u_seq stands for UNIX sequenced packet socket. u_dgr stands for UNIX datagram socket (UDP). u_str is for stream socket (TCP).
    - **Peer Address:** Other end of a connection.
    - **ESTAB:** Stands for established i.e., the connection is active.

```
ubuntuuser@ubuntu25:~$ ss -tuln
Netid         State          Recv-Q         Send-Q                                     Local Address:Port                 Peer Address:Port
udp           UNCONN         0              0                                              127.0.0.54:53                      0.0.0.0:*
udp           UNCONN         0              0                                          127.0.0.53%lo:53                       0.0.0.0:*
udp           UNCONN         0              0                                                0.0.0.0:41177                    0.0.0.0:*
udp           UNCONN         0              0                                              127.0.0.1:323                      0.0.0.0:*
udp           UNCONN         0              0                                                0.0.0.0:45475                    0.0.0.0:*
udp           UNCONN         0              0                                                0.0.0.0:60112                    0.0.0.0:*
udp           UNCONN         0              0                                                0.0.0.0:5353                     0.0.0.0:*
udp           UNCONN         0              0                                                0.0.0.0:52498                    0.0.0.0:*
udp           UNCONN         0              0                                                0.0.0.0:56613                    0.0.0.0:*
udp           UNCONN         0              0                                             10.0.2.15:3702                      0.0.0.0:*
udp           UNCONN         0              0                                      239.255.255.250:3702                       0.0.0.0:*
udp           UNCONN         0              0                                                [::1]:323                          [::]:*
udp           UNCONN         0              0                                                   *:55753                            *:*
udp           UNCONN         0              0                                               [::]:41774                          [::]:*
udp           UNCONN         0              0                                               [::]:5353                           [::]:*
udp           UNCONN         0              0                   [fe80::a00:27ff:fe97:7c11]%enp0s3:3702                          [::]:*
udp           UNCONN         0              0                                     [ff02::c]%enp0s3:3702                         [::]:*
tcp           LISTEN         0              4096                                           127.0.0.54:53                      0.0.0.0:*
tcp           LISTEN         0              4096                                       127.0.0.53%lo:53                       0.0.0.0:*
tcp           LISTEN         0              4096                                           127.0.0.1:631                      0.0.0.0:*
tcp           LISTEN         0              4096                                               [::1]:631                          [::]:*
ubuntuuser@ubuntu25:~$ ss -tn
State              Recv-Q             Send-Q                       Local Address:Port                         Peer Address:Port
ESTAB              0                  0                                10.0.2.15:37774                        142.250.77.106:443
ESTAB              0                  0                                10.0.2.15:56412                        172.64.148.235:443
ESTAB              0                  0                                10.0.2.15:41522                        142.250.77.106:443
ESTAB              0                  0                                10.0.2.15:33334                        172.64.155.209:443
ESTAB              0                  0                                10.0.2.15:33850                       142.251.222.133:443
ESTAB              0                  0                                10.0.2.15:54078                          142.250.66.3:443
ESTAB              0                  0                                10.0.2.15:60330                        172.217.194.84:443
ESTAB              0                  0                                10.0.2.15:42550                        172.64.155.209:443
ESTAB              0                  0                                10.0.2.15:52798                         34.107.243.93:443
ESTAB              0                  0                                10.0.2.15:41932                       142.250.183.170:443
ESTAB              0                  0                                10.0.2.15:41936                       142.250.183.170:443
ESTAB              0                  0                                10.0.2.15:57990                        34.120.208.123:443
ESTAB              0                  0                                10.0.2.15:32796                         34.36.137.203:443
```

- **ifconfig:** Shows IP address, MAC address, network status, and interface statistics. Mainly used to enable/disable network interface.

- **ifconfig interface_name:** To display the specific interface detail only. Eg: **ifconfig enp0s3** will only show information about the main ethernet network interface.
- **ifconfig interface_name up:** Brings interface up.
- **ifconfig interface_name down:** Brings interface down.

```
ubuntuuser@ubuntu25:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fe97:7c11  prefixlen 64  scopeid 0x20<link>
        inet6 fd17:625c:f037:2:a00:27ff:fe97:7c11  prefixlen 64  scopeid 0x0<global>
        ether 08:00:27:97:7c:11  txqueuelen 1000  (Ethernet)
        RX packets 123290  bytes 101991824 (101.9 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 64478  bytes 36056416 (36.0 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

ubuntuuser@ubuntu25:~$ sudo ifconfig lo up
ubuntuuser@ubuntu25:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fe97:7c11  prefixlen 64  scopeid 0x20<link>
        inet6 fd17:625c:f037:2:a00:27ff:fe97:7c11  prefixlen 64  scopeid 0x0<global>
        ether 08:00:27:97:7c:11  txqueuelen 1000  (Ethernet)
        RX packets 123367  bytes 102021152 (102.0 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 64546  bytes 36077639 (36.0 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 5291  bytes 529726 (529.7 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 5291  bytes 529726 (529.7 KB)
        TX errors 0  dropped 134 overruns 0  carrier 0  collisions 0
```

- **traceroute:** Used to trace the path taken by packets from the source to the destination. Helps in identifying network delays or routing issues. By default there are a maximum of 30 hops. A hop is one step a data packet takes from one network device to the next while traveling from a source to a destination.
  - **traceroute google.com:** This commands traces the network path from the system to google.com. Instead of the name IPcan also be given i.e., **traceroute 8.8.8.8 .**

```
ubuntuuser@ubuntu25:~$ traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1  _gateway (10.0.2.2)  0.859 ms  0.386 ms  0.523 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```

- * * * indicates that the router did not respond in all other hops except the first hop.
- **traceroute -m 10 google.com:** Traces the network path but limits the trace to only 10 hops.

```
ubuntuuser@ubuntu25:~$ traceroute -m 10 google.com
traceroute to google.com (142.251.221.174), 10 hops max, 60 byte packets
 1  _gateway (10.0.2.2)  1.222 ms  0.329 ms  2.176 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
ubuntuuser@ubuntu25:~$
```

- **ps:** Stands for process status . ps shows only the processes running in your current terminal (TTY). ps only shows the processes that are currently running in that particular terminal. It is more like a snapshot.
  - **ps** supports multiple option styles. System-V supports a single dash (- i.e., ps -ef). BSD style has no dash (ps aux/ps lax). GNU style uses double dash(ps --tree, ps --all).

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ ps
    PID TTY          TIME CMD
   3447 pts/0    00:00:00 bash
   4150 pts/0    00:00:00 ps
```

  - **Terms:**
    - PID stands for process ID. PID is unique for every process.
    - TTY refers to the current terminal associated with the process.
    - TIME refers to the CPU TIME used by the process.
    - CMD is the command that is running as part of that process.
  - **ps -ef:** show detailed information on every processes that are currently running i.e., the system process as well as user process (processes run by the user).
    - It is a System-V style.

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 16:16 ?        00:00:02 /usr/lib/systemd/systemd --switched-root --system --deserialize=48 splash noprompt noshell automat
root           2       0  0 16:16 ?        00:00:00 [kthreadd]
root           3       2  0 16:16 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 16:16 ?        00:00:00 [kworker/R-rcu_gp]
root           5       2  0 16:16 ?        00:00:00 [kworker/R-sync_wq]
root           6       2  0 16:16 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root           7       2  0 16:16 ?        00:00:00 [kworker/R-slub_flushwq]
root           8       2  0 16:16 ?        00:00:00 [kworker/R-netns]
root          12       2  0 16:16 ?        00:00:00 [kworker/u8:0-ipv6_addrconf]
root          13       2  0 16:16 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14       2  0 16:16 ?        00:00:00 [ksoftirqd/0]
root          15       2  0 16:16 ?        00:00:02 [rcu_preempt]
root          16       2  0 16:16 ?        00:00:00 [rcu_exp_par_gp_kthread_worker/0]
```

  - **Terms:**
    - **UID:** User ID owning the process
    - **PID:** Process ID

- **PPID:** Parent process ID. The parent process is the process which resulted in creating the new process. PPID 0 means no parent.
- **C:** Represents the total CPU usage.
- **STIME:** Shows when the process was started.
- **TTY:** Refers to the terminal in which the command was executed. '**?**' implies there is no terminal.

- **ps aux:** BSD style. No dash is needed. Displays all running processes, including background daemons, with CPU and memory usage. More user friendly.
  - a: Show processes of all users with terminal
  - u: User-oriented format
  - x: Include processes without a controlling terminal (daemons)

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1  25172  6428 ?        Ss   16:16   0:02 /usr/lib/systemd/systemd --switched-root --system --deserialize=48 splash noprompt
root           2  0.0  0.0      0     0 ?        S    16:16   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    16:16   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   16:16   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   16:16   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   16:16   0:00 [kworker/R-kvfree_rcu_reclaim]
root           7  0.0  0.0      0     0 ?        I<   16:16   0:00 [kworker/R-slub_flushwq]
root           8  0.0  0.0      0     0 ?        I<   16:16   0:00 [kworker/R-netns]
root          12  0.0  0.0      0     0 ?        I    16:16   0:00 [kworker/u8:0-ipv6_addrconf]
root          13  0.0  0.0      0     0 ?        I<   16:16   0:00 [kworker/R-mm_percpu_wq]
root          14  0.0  0.0      0     0 ?        S    16:16   0:00 [ksoftirqd/0]
root          15  0.1  0.0      0     0 ?        I    16:16   0:03 [rcu_preempt]
root          16  0.0  0.0      0     0 ?        S    16:16   0:00 [rcu_exp_par_gp_kthread_worker/0]
root          17  0.0  0.0      0     0 ?        S    16:16   0:00 [rcu_exp_gp_kthread_worker]
root          18  0.0  0.0      0     0 ?        S    16:16   0:00 [migration/0]
root          19  0.0  0.0      0     0 ?        S    16:16   0:00 [idle_inject/0]
root          20  0.0  0.0      0     0 ?        S    16:16   0:00 [cpuhp/0]
root          21  0.0  0.0      0     0 ?        S    16:16   0:00 [cpuhp/1]
root          22  0.0  0.0      0     0 ?        S    16:16   0:00 [idle_inject/1]
root          23  0.0  0.0      0     0 ?        S    16:16   0:00 [migration/1]
root          24  0.0  0.0      0     0 ?        S    16:16   0:02 [ksoftirqd/1]
root          27  0.1  0.0      0     0 ?        I    16:16   0:05 [kworker/u9:0-events_power_efficient]
root          28  0.1  0.0      0     0 ?        I    16:16   0:05 [kworker/u10:0-events_unbound]
root          29  0.0  0.0      0     0 ?        S    16:16   0:00 [kdevtmpfs]
root          30  0.0  0.0      0     0 ?        I<   16:16   0:00 [kworker/R-inet_frag_wq]
root          31  0.0  0.0      0     0 ?        I    16:16   0:00 [rcu_tasks_kthread]
root          32  0.0  0.0      0     0 ?        I    16:16   0:00 [rcu_tasks_rude_kthread]
root          33  0.0  0.0      0     0 ?        I    16:16   0:00 [rcu_tasks_trace_kthread]
root          34  0.0  0.0      0     0 ?        S    16:16   0:00 [kauditd]
root          35  0.0  0.0      0     0 ?        S    16:16   0:00 [khungtaskd]
root          36  0.0  0.0      0     0 ?        S    16:16   0:00 [oom_reaper]
```

- **Terms:**
  - **%CPU:** CPU usage percentage.
  - **%MEM:** Memory usage percentage**.**
  - **VSZ:** Virtual memory size in KB.
  - **RSS:** Physical RAM used in KB.
  - **STAT:** Indicates the **Process States**. It can be:
    - R: Currently running or ready to run
    - S: Sleeping. Waiting for an event and it can be interrupted.
    - T: Process execution is stopped or paused.
    - Z: Zombie process. Process that has finished execution but still exists in the process table.

- **ps -l:** Long detailed list of processes.

```
ubuntuuser@ubuntu25:~$ ps -l
F S   UID     PID    PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
0 S  1000    3447    3384  0  80   0 -  5172 do_wai pts/0    00:00:00 bash
4 R  1000    7231    3447  9  80   0 -  5366 -      pts/0    00:00:00 ps
ubuntuuser@ubuntu25:~$
```

- NI is Nice Value and PRI is Kernel Priority.
- It defines priority. If PRI is 0 and NI is -20, then the process has high priority.
- If PRT is 20 and NI is 0 then middle priority.
- If NI is 19, then it is a low priority process.
- SZ is virtual memory size measured in pages.

- Top: real-time system monitoring command. Refreshes every 3 seconds.

```
ubuntuuser@ubuntu25:~$ top

top - 17:41:49 up  1:25,  1 user,  load average: 2.51, 2.09, 1.90
Tasks: 230 total,   3 running, 223 sleeping,   0 stopped,   4 zombie
%Cpu(s):  2.2 us, 38.8 sy,  0.0 ni, 37.6 id, 10.4 wa,  0.0 hi, 11.0 si,  0.0 st
MiB Mem :   3402.1 total,    117.2 free,   3162.0 used,    426.3 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    240.1 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   7430 ubuntuu+  20   0 3482740  61028  43768 S  35.6   1.8   0:01.08 Isolated Servic
   3467 ubuntuu+  20   0   12.5g 358780  66004 S  21.8  10.3  16:27.05 firefox
     56 root      20   0       0      0      0 S  11.2   0.0   0:21.05 kswapd0
   2213 ubuntuu+  20   0 4100512 348940  64232 R   7.9  10.0  17:18.45 gnome-shell
   7118 root      20   0       0      0      0 I   5.6   0.0   0:03.16 kworker/u10:2-loop14
   6912 root      20   0       0      0      0 I   4.6   0.0   0:01.85 kworker/u10:4-loop6
   3337 ubuntuu+  20   0 2794012  27432  12816 S   4.3   0.8   0:02.96 gjs
   7069 root      20   0       0      0      0 I   4.3   0.0   0:00.79 kworker/u9:1-loop6
   4424 ubuntuu+  20   0 4226276 490016  28196 S   3.3  14.1   1:00.97 Isolated Web Co
   3376 ubuntuu+  20   0 1657180 130024  19316 R   1.7   3.7   0:14.19 ptyxis
   6771 root      20   0       0      0      0 I   1.3   0.0   0:03.43 kworker/u9:4-writeback
   7429 ubuntuu+  20   0   22216   3588   1408 R   1.3   0.1   0:00.08 top
     14 root      20   0       0      0      0 S   0.7   0.0   0:01.81 ksoftirqd/0
     24 root      20   0       0      0      0 S   0.7   0.0   0:03.47 ksoftirqd/1
    374 root      19  -1   68016   4588   2828 S   0.7   0.1   0:03.17 systemd-journal
   6561 root       0 -20       0      0      0 I   0.7   0.0   0:01.26 kworker/1:0H-kblockd
   7251 root       0 -20       0      0      0 I   0.7   0.0   0:00.30 kworker/0:0H-kblockd
     15 root      20   0       0      0      0 I   0.3   0.0   0:06.91 rcu_preempt
```

- **htop:** htop is an interactive, real-time process viewer for Linux, similar to top but much more user-friendly and visually informative.
  - **Common Actions:**
    - **F2:** Setup / customize columns and display.
    - **F3:** Search for a process by name.
    - **F4:** Filter processes.
    - **F5:** Tree view.
    - **F6:** Sort on the basis of certain values. It can be Priority, Nice value, State etc.
    - **F7/F8:** Increase or decrease the priority.
    - **F9:** Kill a process.
    - **F10:** Exit/ quit htop interface.

```
  0[|||||                                                    5.5%] Tasks: 140, 621 thr, 85 kthr; 2 running
  1[||                                                       2.1%] Load average: 1.31 1.06 1.09
Mem[||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||2.56G/3.32G] Uptime: 01:59:35
Swp[                                                        0K/0K]

 Main   I/O
   PID USER      PRI△ NI  VIRT   RES   SHR S  CPU% MEM%   TIME+  Command
  1554 rtkit      RT   1 21900  2112     0 S   0.0  0.1  0:00.04 /usr/libexec/rtkit-daemon
  2043 ubuntuuser -21   0  111M 10744     0 S   0.0  0.3  0:00.70 /usr/bin/pipewire
  2062 ubuntuuser -21   0  188M 19348     0 S   0.0  0.6  0:00.05 /usr/bin/pipewire-pulse
  2079 ubuntuuser -21   0  479M  9980     0 S   0.0  0.3  0:00.00 /usr/bin/wireplumber
  7923 ubuntuuser -11   0 12.6G  495M     0 S   0.0 14.6  0:00.00 /snap/firefox/7477/usr/lib/firefox/firefox
  2229 ubuntuuser   5 -15 4001M  371M     0 S   1.1 10.9  0:40.48 /usr/bin/gnome-shell
  2024 ubuntuuser   9 -11  111M 10744  5996 S   0.0  0.3  0:00.18 /usr/bin/pipewire
  2031 ubuntuuser   9 -11  479M  9980  5008 S   0.0  0.3  0:00.49 /usr/bin/wireplumber
  2032 ubuntuuser   9 -11  188M 19348  7316 S   0.0  0.6  0:00.27 /usr/bin/pipewire-pulse
   374 root        19  -1 68016 10984  9220 S   0.0  0.3  0:03.73 /usr/lib/systemd/systemd-journald
     1 root        20   0 25172  9952  4880 S   0.0  0.3  0:02.96 /usr/lib/systemd/systemd --switched-root --system --deserialize=48 splash noprompt
   427 systemd-oo  20   0 18300  2904  1876 S   0.0  0.1  0:02.60 /usr/lib/systemd/systemd-oomd
   428 systemd-re  20   0 23536  5828  2792 S   0.0  0.2  0:02.11 /usr/lib/systemd/systemd-resolved
   437 root        20   0 38844  5348  1064 S   0.0  0.2  0:00.30 /usr/lib/systemd/systemd-udevd
  1118 avahi       20   0  6780  1072   680 S   0.0  0.0  0:00.20 avahi-daemon: running [ubuntu25.local]
  1119 root        20   0  3040   240   124 S   0.0  0.0  0:00.03 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
  1120 messagebus  20   0 11960  5660  1728 S   0.0  0.2  0:01.69 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation
  1130 root        20   0 52884 14708   500 S   0.0  0.4  0:00.17 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
  1141 polkitd     20   0  388M  8760  5064 S   0.0  0.3  0:00.33 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
  1144 root        20   0 1878M 21820  5684 S   0.0  0.6  0:00.79 /snap/snapd/current/usr/lib/snapd/snapd
  1145 root        20   0  313M  1488   644 S   0.0  0.0  0:00.06 /usr/libexec/accounts-daemon
  1148 root        20   0 10540  1120   872 S   0.0  0.0  0:00.17 /usr/sbin/cron -f -P
  1151 root        20   0  311M  1164   524 S   0.0  0.0  0:00.02 /usr/libexec/switcheroo-control
  1154 root        20   0 19064  4312  3008 S   0.0  0.1  0:00.36 /usr/lib/systemd/systemd-logind
  1155 root        20   0  459M  5912  3104 S   0.0  0.2  0:00.14 /usr/libexec/udisks2/udisksd
  1172 root        20   0  311M  1164     0 S   0.0  0.0  0:00.00 /usr/libexec/switcheroo-control
  1173 root        20   0  311M  1164     0 S   0.0  0.0  0:00.00 /usr/libexec/switcheroo-control
  1177 syslog      20   0  215M  2564   940 S   0.0  0.1  0:00.05 /usr/sbin/rsyslogd -n -iNONE
  1178 root        20   0  459M  5912     0 S   0.0  0.2  0:00.16 /usr/libexec/udisks2/udisksd
  1179 root        20   0  459M  5912     0 S   0.0  0.2  0:00.00 /usr/libexec/udisks2/udisksd
F1Help  F2Setup F3SearchF4FilterF5Tree  F6SortByF7Nice -F8Nice +F9Kill  F10Quit
```

- **iotop:** It shows which processes are using disk or swap I/O, how much, and in what order. It's like the top command but for disk activity instead of CPU/memory usage.Root privileges are required to monitor I/O of all process.

```
Total DISK READ:         0.00 B/s | Total DISK WRITE:         0.00 B/s
Current DISK READ:       0.00 B/s | Current DISK WRITE:       0.00 B/s
    TID  PRIO  USER     DISK READ DISK WRITE>    COMMAND
      1 be/4 root        0.00 B/s    0.00 B/s systemd --switched-root --system --deserialize=48 s~4G-32G:512M,32G-64G:1024M,64G-128G:2048M,128G-:4096
      2 be/4 root        0.00 B/s    0.00 B/s [kthreadd]
      3 be/4 root        0.00 B/s    0.00 B/s [pool_workqueue_release]
      4 be/0 root        0.00 B/s    0.00 B/s [kworker/R-rcu_gp]
      5 be/0 root        0.00 B/s    0.00 B/s [kworker/R-sync_wq]
      6 be/0 root        0.00 B/s    0.00 B/s [kworker/R-kvfree_rcu_reclaim]
      7 be/0 root        0.00 B/s    0.00 B/s [kworker/R-slub_flushwq]
      8 be/0 root        0.00 B/s    0.00 B/s [kworker/R-netns]
     12 be/4 root        0.00 B/s    0.00 B/s [kworker/u8:0-ipv6_addrconf]
     13 be/0 root        0.00 B/s    0.00 B/s [kworker/R-mm_percpu_wq]
     14 be/4 root        0.00 B/s    0.00 B/s [ksoftirqd/0]
     15 be/4 root        0.00 B/s    0.00 B/s [rcu_preempt]
     16 be/4 root        0.00 B/s    0.00 B/s [rcu_exp_par_gp_kthread_worker/0]
     17 be/4 root        0.00 B/s    0.00 B/s [rcu_exp_gp_kthread_worker]
     18 rt/4 root        0.00 B/s    0.00 B/s [migration/0]
     19 rt/4 root        0.00 B/s    0.00 B/s [idle_inject/0]
     20 be/4 root        0.00 B/s    0.00 B/s [cpuhp/0]
     21 be/4 root        0.00 B/s    0.00 B/s [cpuhp/1]
     22 rt/4 root        0.00 B/s    0.00 B/s [idle_inject/1]
     23 rt/4 root        0.00 B/s    0.00 B/s [migration/1]
     24 be/4 root        0.00 B/s    0.00 B/s [ksoftirqd/1]
     29 be/4 root        0.00 B/s    0.00 B/s [kdevtmpfs]
     30 be/0 root        0.00 B/s    0.00 B/s [kworker/R-inet_frag_wq]
     31 be/4 root        0.00 B/s    0.00 B/s [rcu_tasks_kthread]
     32 be/4 root        0.00 B/s    0.00 B/s [rcu_tasks_rude_kthread]
     33 be/4 root        0.00 B/s    0.00 B/s [rcu_tasks_trace_kthread]
     34 be/4 root        0.00 B/s    0.00 B/s [kauditd]
     35 be/4 root        0.00 B/s    0.00 B/s [khungtaskd]
     36 be/4 root        0.00 B/s    0.00 B/s [oom_reaper]
     38 be/0 root        0.00 B/s    0.00 B/s [kworker/R-writeback]
     39 be/4 root        0.00 B/s    0.00 B/s [kcompactd0]
     40 be/5 root        0.00 B/s    0.00 B/s [ksmd]
     42 be/7 root        0.00 B/s    0.00 B/s [khugepaged]
  keys:  any: refresh  q: quit  i: ionice  o: active  p: procs  a: accum
  sort:  r: asc  left: DISK READ  right: COMMAND  home: TID  end: COMMAND
CONFIG_TASK_DELAY_ACCT and kernel.task_delayacct sysctl not enabled in kernel, cannot determine SWAPIN and IO %
```

  - **Terms:**
    - **DISK READ:** Current disk read rate
    - **DISK WRITE:** Current disk write rate
    - **TID:** Thread ID

        - **PID:** Process ID
    - **Commands Used:**
        - **any:** Press any key to refresh
        - q: Quit
        - i: Toggle display of I/O priority (ionice) column. It is like the nice value for disk I/O. Allows you to see which processes have higher/lower I/O priority.
        - p: Toggle between showing process only or threads.
        - r: Reverse sorting (Descending order).
        - left/right arrow: Move cursor to left or right column to sort on the basis of the selected column.
        - home: Move focus to PID which is the first column to sort on the basis of PID.
        - end: Go to the last column which is COMMAND for sorting.

- **awk**
    - Tool to read files line by line, split each line into fields (columns), and perform actions.
    - awk 'pattern { action }' filename
    - Fields are indicated using $field number (eg: $2) to indicate the field associated with the number.
    - OFS : Output Field Separator. Used to separate each field using symbols or characters enclosed in double quotes.

```
ubuntuuser@ubuntu25:~/Documents/csttasks/awk$ awk '{ OFS=":"; print $1, $3, $5 }' employee.txt
101:25:50000
102:30:60000
103:22:35000
104:28:55000
ubuntuuser@ubuntu25:~/Documents/csttasks/awk$ awk '{ OFS="-"; print $1, $3, $5 }' employee.txt
101-25-50000
102-30-60000
103-22-35000
104-28-55000
```

    - awk can be combined with Operators, Conditional statements, Loops etc.
    - awk -v is used to set a user- defined variable.

```
ubuntuuser@ubuntu25:~/Documents/csttasks/awk$ cat employee.txt
101 John 25 Developer 50000
102 Alice 30 Designer 60000
103 Bob 22 Tester 35000
104 Eve 28 Developer 55000
ubuntuuser@ubuntu25:~/Documents/csttasks/awk$ awk '{ print $2, $4 }' employee.txt
John Developer
Alice Designer
Bob Tester
Eve Developer
ubuntuuser@ubuntu25:~/Documents/csttasks/awk$ awk '{ print $NF }' employee.txt
50000
60000
35000
55000
ubuntuuser@ubuntu25:~/Documents/csttasks/awk$ awk '{ print NR, $2, $4 }' employee.txt
1 John Developer
2 Alice Designer
3 Bob Tester
4 Eve Developer
ubuntuuser@ubuntu25:~/Documents/csttasks/awk$ awk '$5 > 50000 { print $2, $4, $5 }' employee.txt
Alice Designer 60000
Eve Developer 55000
ubuntuuser@ubuntu25:~/Documents/csttasks/awk$ awk '$5 > 50000 && $4 == "Developer" { print $2, $4, $5 }' employee.txt
Eve Developer 55000
ubuntuuser@ubuntu25:~/Documents/csttasks/awk$ awk '$4 == "Tester" || $4 == "Developer" { print $2, $4, $5 }' employee.txt
John Developer 50000
Bob Tester 35000
Eve Developer 55000
ubuntuuser@ubuntu25:~/Documents/csttasks/awk$ awk -v min_age=23 '$3 > min_age { print $2, $3 }' employee.txt
John 25
Alice 30
Eve 28
```

- **sed**
  - Text editor used to search, find, replace, insert, delete, and modify text in files or input streams.
  - sed ‘command’ filename
  - **sed 's/pattern/replacement/' file:** To search and replace
  - **sed '/pattern/d' file:** To delete the line containing the pattern
  - **sed 'Nd' file:** Delete a specific line N.
  - **sed -n 'Np' file:** Prints the given number if lines
  - **sed -n 'N,Mp' file:** Prints specific range of lines.
  - **sed 'Ni\text' file:** Inserts a line of text before a line N.

```
ubuntuuser@ubuntu25:~/Documents/csttasks/sed$ cat sample.txt
Linux is a powerful operating system.
It is widely used in servers and cloud platforms.
Linux provides strong security and flexibility.
Many developers prefer Linux for programming.

ubuntuuser@ubuntu25:~/Documents/csttasks/sed$ sed 's/Linux/Unix/g' sample.txt
Unix is a powerful operating system.
It is widely used in servers and cloud platforms.
Unix provides strong security and flexibility.
Many developers prefer Unix for programming.

ubuntuuser@ubuntu25:~/Documents/csttasks/sed$ sed '/Many/d' sample.txt
Linux is a powerful operating system.
It is widely used in servers and cloud platforms.
Linux provides strong security and flexibility.

ubuntuuser@ubuntu25:~/Documents/csttasks/sed$ sed '2d' sample.txt
Linux is a powerful operating system.
Linux provides strong security and flexibility.
Many developers prefer Linux for programming.

ubuntuuser@ubuntu25:~/Documents/csttasks/sed$ sed -n '1p' sample.txt
Linux is a powerful operating system.
ubuntuuser@ubuntu25:~/Documents/csttasks/sed$ sed -n '1,3p' sample.txt
Linux is a powerful operating system.
It is widely used in servers and cloud platforms.
Linux provides strong security and flexibility.
ubuntuuser@ubuntu25:~/Documents/csttasks/sed$ sed '3i\There is Two type of license in Linux\' sample.txt
Linux is a powerful operating system.
It is widely used in servers and cloud platforms.
There is Two type of license in Linux
Linux provides strong security and flexibility.
Many developers prefer Linux for programming.
```

- **sed 'Na\text' file:** Append a line after a line N.

```
ubuntuuser@ubuntu25:~/Documents/csttasks/sed$ sed '3a\These Redhat based and Debian based\' sample.txt
Linux is a powerful operating system.
It is widely used in servers and cloud platforms.
Linux provides strong security and flexibility.
These Redhat based and Debian based
Many developers prefer Linux for programming.
```

- These won't change the contents of the file unless it's demanded. For that we use **sed -i** (in place editing) which actually edits the original file.

```
ubuntuuser@ubuntu25:~/Documents/csttasks/sed$ cat sample.txt
Linux is a powerful operating system.
It is widely used in servers and cloud platforms.
Linux provides strong security and flexibility.
Many developers prefer Linux for programming.
ubuntuuser@ubuntu25:~/Documents/csttasks/sed$ sed '3a\These Redhat based and Debian based\' sample.txt
Linux is a powerful operating system.
It is widely used in servers and cloud platforms.
Linux provides strong security and flexibility.
These Redhat based and Debian based
Many developers prefer Linux for programming.
ubuntuuser@ubuntu25:~/Documents/csttasks/sed$ cat sample.txt
Linux is a powerful operating system.
It is widely used in servers and cloud platforms.
Linux provides strong security and flexibility.
Many developers prefer Linux for programming.
ubuntuuser@ubuntu25:~/Documents/csttasks/sed$ sed -i '$a\These are the Basics of Linux\' sample.txt
ubuntuuser@ubuntu25:~/Documents/csttasks/sed$ cat sample.txt
Linux is a powerful operating system.
It is widely used in servers and cloud platforms.
Linux provides strong security and flexibility.
Many developers prefer Linux for programming.
These are the Basics of Linux
ubuntuuser@ubuntu25:~/Documents/csttasks/sed$
```

Day2 - 16/12/2025

Tasks Completed Today:

- **Task1:** Familiarized linux file permissions: chmod command.
- **Task2:** Learned changing file ownership using the chown command.
- **Task3:** Familiarized securing linux server using Uncomplicated firewall (UFW).
- **Task4:** Familiarized firewall-cmd command by adding services, ports etc. to the firewall.
- **Task5:** Automated tasks like checking status of services, changing permissions of files having full permissions, checking system informations, updating database.

Learnings:

- **File Permissions:**
  - File permissions can be viewed most commonly using the command ls -l filename.
  - The output given below shows details of a file named permission.txt.

- Permissions are of three groups: User (u), Group (g), Others (o).
- The permission can be granted for read (r), write (w) and execute(x).
- By default a file has rwx(read, write and execute) for the user, rx (read and execute only) for the group to which the user belongs and rx for others. Ungranted permissions are denoted using dash (-).
- Dash (-) in front of a file list indicates that it is a regular file. It would be denoted using d if it was a directory.
- Permission can be changed using the chmod command.
- '+/-' is used for adding / removing the permission for the specific groups.
- Two methods of changing permissions:

  - **Symbolic method**
    - chmod u+x: gives user executing permission for a file.
    - chmod g-w: removes the write permission for a file.
    - chmod o+rwx: grants others users who are not the owner and not in the file group the permission to read, write and execute a file. More than one permission can be granted or removed all at once.
    - chmod a+rwx: Sets all permission for all type of groups.

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l permission.txt
-rw-rw-r-- 1 ubuntuuser ubuntuuser 0 Dec 16 05:51 permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$ chmod u+x permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l permission.txt
-rwxrw-r-- 1 ubuntuuser ubuntuuser 0 Dec 16 05:51 permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$ chmod u-x permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l permission.txt
-rw-rw-r-- 1 ubuntuuser ubuntuuser 0 Dec 16 05:51 permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$ chmod o+rwx permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l permission.txt
-rw-rw-rwx 1 ubuntuuser ubuntuuser 0 Dec 16 05:51 permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$ chmod g-rw permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l permission.txt
-rw----rwx 1 ubuntuuser ubuntuuser 0 Dec 16 05:51 permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$
```

  - **Numeric Method**
    - Instead of symbols numbers are used to represent file permissions.
    - Values are assigned for read (4), write (2) and execute(1).
    - 4+2+1= 7 which grants all permission to a group.
    - 4+1=5 only grants read and execute permission.

- 4+2=6 indicates read and write permission only.
- 2+1=3 indicates write and execute permission only.
- chmod 777 permission.txt: gives all permission for a file named permission.txt to all groups i.e., user =7 , group = 7 and others =7.
- chmod 730: Grants **full permission (7)** to user , **write and execute (3)** to group and **no permission (0)** to others.

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ chmod 777 permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l permission.txt
-rwxrwxrwx 1 ubuntuuser ubuntuuser 0 Dec 16 05:51 permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$ chmod 730 permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l permission.txt
-rwx-wx--- 1 ubuntuuser ubuntuuser 0 Dec 16 05:51 permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$ chmod 421 permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l permission.txt
-r---w---x 1 ubuntuuser ubuntuuser 0 Dec 16 05:51 permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$
```

- **Special Permissions:**
  - **SUID(Set User ID)**
    - When a file is executed, it runs with the owner's permissions, not the user's. When another user executes a file which has its original user, after executing the file, it is recognised to be executed by the original user even if it was executed by another user.
    - Numeric value = 4 i.e., chmod 4755 filename enables special permission for the user.
    - Symbolic method: chmod u+s filename.. SUID is denoted by 's' in the user group in place of x i.e., execution. If you don't give execution permission for that file special permission will be denoted as 'S' (uppercase) i.e., special permission is granted but not enabled.

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ chmod u+s test.sh
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l test.sh
-rwsr-xr-x 1 ubuntuuser ubuntuuser 22 Dec 16 07:56 test.sh
ubuntuuser@ubuntu25:~/Documents/csttasks$ chmod 4666 test.sh
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l test.sh
-rwSrw-rw- 1 ubuntuuser ubuntuuser 22 Dec 16 07:56 test.sh
ubuntuuser@ubuntu25:~/Documents/csttasks$
```

  - **SGID (Set Group ID):**
    - If a group is given Special permission, then for every file or directory that any user belonging to that group creates will have the group name as group instead of the user name as group.

- Numeric value =7. Indicated by a ‘s’ (lowercase) in the group which means enabled. ‘S’ (uppercase) means special permission for groups is disabled.
- Symbolic method = chmod g+s filename for granting SGID and g-s for removing SGID.

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l permission.txt
-r---w---x 1 ubuntuuser ubuntuuser 0 Dec 16 05:51 permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$ chmod g+s permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l permission.txt
-r---wS--x 1 ubuntuuser ubuntuuser 0 Dec 16 05:51 permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$ chmod 2755 permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l permission.txt
-rwxr-sr-x 1 ubuntuuser ubuntuuser 0 Dec 16 05:51 permission.txt
ubuntuuser@ubuntu25:~/Documents/csttasks$
```

- **Sticky bit:**
  - Denoted by ‘t’ if enabled and ‘T’ if disabled.
  - If sticky bit is enabled for a directories no other user belonging to the same group or different group can delete the file.
  - Useful in case of shared directories.
  - Only the file owner and the directory owner can delete the file.
  - Numeric value = 1. chmod 1777 directory name.
  - Symbolic method:
    - chmod +t dir_name: for enabling
    - chmod -t dir_name: for disabling

- **UFW (Uncomplicated firewall)**
  - UFW (Uncomplicated Firewall) is a simple and user-friendly firewall management tool used on Linux systems to control network traffic.
  - Sudo privilege is needed to run ufw.
  - **ufw status:** command used to check the current state of the UFW i.e., whether it is active or not and to view the firewall rules that are currently applied on your system.
  - Default security rules can be set:
    - **ufw default deny incoming:** This will block all incoming connections.
    - **ufw default allow outcoming:** This will access to outoginh connections.

- **ufw allow ssh:** Essential services can be allowed by specifying name or port . eg: **ufw allow 22/tcp.** Allowing the same command several times will create duplicate rule.

```
root@ubuntu25:~# ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
root@ubuntu25:~# ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
root@ubuntu25:~# ufw allow ssh
Rules updated
Rules updated (v6)
root@ubuntu25:~# ufw allow 22/tcp
Skipping adding existing rule
Skipping adding existing rule (v6)
root@ubuntu25:~# ufw allow 80/tcp
Rules updated
Rules updated (v6)
root@ubuntu25:~# ufw allow 443/tcp
Rules updated
Rules updated (v6)
root@ubuntu25:~# ufw allow https
Rules updated
Rules updated (v6)
root@ubuntu25:~# ufw enable
Firewall is active and enabled on system startup
root@ubuntu25:~# ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
80/tcp                     ALLOW       Anywhere
443/tcp                    ALLOW       Anywhere
443                        ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
80/tcp (v6)                ALLOW       Anywhere (v6)
443/tcp (v6)               ALLOW       Anywhere (v6)
443 (v6)                   ALLOW       Anywhere (v6)
```

- **ufw deny 192.168.56.1:** Connections from specific IPs can be allowed or denied.
- **ufw allow 192.168.28.9 to any port 22:** Connection from any source IP to any specific port only on the host server can be allowed or denied.
- **ufw enable:** Firewall is active and enabled on system startup.
- ufw status numbered will show existing rules that are allowed in with numbers assigned.
- **ufw delete number:** Deletes the with the assigned number.
- Ufw disable: Stops firewall

```
root@ubuntu25:~# ufw status numbered
Status: active

     To                         Action      From
     --                         ------      ----
[ 1] 22/tcp                     ALLOW IN    Anywhere
[ 2] 80/tcp                     ALLOW IN    Anywhere
[ 3] 443/tcp                    ALLOW IN    Anywhere
[ 4] 443                        ALLOW IN    Anywhere
[ 5] Anywhere                   ALLOW IN    192.168.1.10
[ 6] 22/tcp (v6)                ALLOW IN    Anywhere (v6)
[ 7] 443/tcp (v6)               ALLOW IN    Anywhere (v6)

root@ubuntu25:~# ufw delete 7
Deleting:
 allow 443/tcp
Proceed with operation (y|n)? y
Rule deleted (v6)
root@ubuntu25:~# ufw disable
Firewall stopped and disabled on system startup
root@ubuntu25:~#
```

- **chown:** to change ownership of a file. The command changes ownership of a file or a directory to a specific user in a specific group.

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l task
-rw-rw-r-- 1 ubuntuuser ubuntuuser 0 Dec 12 08:44 task
ubuntuuser@ubuntu25:~/Documents/csttasks$ sudo chown a:cst task
[sudo: authenticate] Password:
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l task
-rw-rw-r-- 1 a cst 0 Dec 12 08:44 task
ubuntuuser@ubuntu25:~/Documents/csttasks$
```

- **chgrp:** This command changes the group of a file or directory from the existing group.

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l file3
-rw-rw-r-- 1 ubuntuuser ubuntuuser 0 Dec 12 08:00 file3
ubuntuuser@ubuntu25:~/Documents/csttasks$ sudo chgrp cst file3
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls -l file3
-rw-rw-r-- 1 ubuntuuser cst 0 Dec 12 08:00 file3
ubuntuuser@ubuntu25:~/Documents/csttasks$
```

- **Automate Daily Tasks:**
  - For automating Daily like checking status of services, changing permissions of files having full permissions, checking system information, updating database etc. A shell script is written to run all these tasks together.

```bash
#!/bin/bash
echo "=== Service Status ==="
SERVICES=("ufw" "ssh" "mysql")
for service in "${SERVICES[@]}"
do
        status=$(systemctl is-active $service)
        echo "$service : $status"
done

echo "=== Fixing Insecure File Permision to ==="
find /home -type f -perm 0777 -exec chmod 0644{} \;
echo "=== Completed ==="

echo "=== System Information ==="
uptime
df -h
free -h
uname -a

echo "=== Updating System Packages ==="
apt update -y && apt upgrade -y
```

- Test if the script is running properly

```
root@ubuntu25:~# ./automation.sh
=== Service Status ===
ufw : active
ssh : inactive
mysql : inactive
=== Fixing Insecure File Permision to ===
=== Completed ===
=== System Information ===
 12:08:19 up  1:56,  2 users,  load average: 0.90, 1.53, 1.02
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           681M  1.8M  679M   1% /run
/dev/sda2        30G  7.0G   21G  25% /
tmpfs           1.7G     0  1.7G   0% /dev/shm
tmpfs           5.0M  8.0K  5.0M   1% /run/lock
tmpfs           1.7G  1.5M  1.7G   1% /tmp
tmpfs           1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
tmpfs           341M   92K  341M   1% /run/user/1000
tmpfs           341M   48K  341M   1% /run/user/0
               total        used        free      shared  buff/cache   available
Mem:           3.3Gi       3.0Gi       112Mi       135Mi       527Mi       321Mi
Swap:             0B          0B          0B
Linux ubuntu25 6.17.0-8-generic #8-Ubuntu SMP PREEMPT_DYNAMIC Fri Nov 14 21:44:46 UTC 2025 x86_64 GNU/Linux
=== Updating System Packages ===
Hit:1 http://security.ubuntu.com/ubuntu questing-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu questing InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu questing-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu questing-backports InRelease
All packages are up to date.
The following packages were automatically installed and are no longer required:
  linux-headers-6.17.0-5  linux-headers-6.17.0-5-generic  linux-modules-6.17.0-5-generic  linux-tools-6.17.0-5  linux-tools-6.17.0-5-generic
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
```

- Create a cron job to execute this script at a specific time. Here **crontab -e** is the command used to create a cron job. After creating, verify the cronjob by **crontab -l.**

```
root@ubuntu25:~# crontab -e
crontab: installing new crontab
root@ubuntu25:~# crontab -l

0 18 * * * /root/automation.sh >> /var/log/automation.log 2>&1
root@ubuntu25:~#
```

- This is a cron job that will schedule the execution of the script at 6 PM everyday.
- At 6 P M the cron job will execute the automation script and the output will be stored in automation.log.

```
root@ubuntu25:~# cat /var/log/automation.log
=== Service Status ===
ufw : active
ssh : inactive
mysql : inactive
=== Fixing Insecure File Permision to ===
=== Completed ===
=== System Information ===
 18:00:02 up 14 min,  2 users,  load average: 3.65, 2.47, 1.23
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           681M  1.7M  679M   1% /run
/dev/sda2        30G  7.0G   21G  26% /
tmpfs           1.7G     0  1.7G   0% /dev/shm
tmpfs           5.0M  8.0K  5.0M   1% /run/lock
tmpfs           1.7G  8.0K  1.7G   1% /tmp
tmpfs           1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
tmpfs           341M   88K  341M   1% /run/user/1000
tmpfs           341M   48K  341M   1% /run/user/0
               total        used        free      shared  buff/cache   available
Mem:           3.3Gi       3.0Gi       113Mi       161Mi       495Mi       316Mi
Swap:             0B          0B          0B
Linux ubuntu25 6.17.0-8-generic #8-Ubuntu SMP PREEMPT_DYNAMIC Fri Nov 14 21:44:46 UTC 2025 x86_64 GNU/Linux
=== Updating System Packages ===

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

Hit:1 http://in.archive.ubuntu.com/ubuntu questing InRelease
Hit:2 http://security.ubuntu.com/ubuntu questing-security InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu questing-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu questing-backports InRelease
Reading package lists...
Building dependency tree...
Reading state information...
All packages are up to date.

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.
```

Day 3 - 17/12/2025

Tasks Completed Today:

- **Task1:** Executed shell scripts using conditional statements and arithmetic, logic and relational operators.
- **Task2:** Wrote shell script to backup the files given as argument with file name current date and time. Executed in terminal and tested the results.

Learning:

- **Conditional Statements:**
  - **if:** used for single conditions.
    - **Syntax:**

- if [condition]; then
  commands
  fi

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ cat if.sh
#!/bin/bash
echo "Enter the Number:"
read a
if [ $a -gt 5 ]; then
        echo "$a is greater than 5"
fi


ubuntuuser@ubuntu25:~/Documents/csttasks$ ./if.sh
Enter the Number:
10
10 is greater than 5
```

- **if-else:** Used if there are alternate conditions.
  - **Syntax:**
    - if [condition]
      then
        # Executes if condition is true
      else
        # Executes if condition is false
      fi

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ cat else.sh
#1/bin/bash
echo "Enter First number"
read a
echo "Enter Second number"
read b
if [ $a -ge $b ]
then
        echo "$a is greater than or equal to $b"

else [ $a -le $b ]
        echo "$a is less than or equal to $b"
fi
ubuntuuser@ubuntu25:~/Documents/csttasks$ ./else.sh
Enter First number
4
Enter Second number
8
4 is less than or equal to 8
ubuntuuser@ubuntu25:~/Documents/csttasks$ ./else.sh
Enter First number
10
Enter Second number
2
10 is greater than or equal to 2
```

- **elif:** Used for multiple conditions

- **Syntax:**
  - if [condition1]; then
    echo "Condition 1 is true"
    elif [condition2]; then
    echo "Condition 2 is true"
    elif [condition3]; then
    echo "Condition 3 is true"
    else
    echo "None of the above"
    fi

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ cat elif.sh
#!/bin/bash
echo "Enter number a"
read a
echo "Enter number b"
read b
if [ $a -gt $b ]
then
        echo "$a is greater than $b"
elif [ $a -lt $b ]
then
        echo "$a is less than $b"
elif [ $a -eq $b ]
then
        echo "$a is equal to $b"
fi



ubuntuuser@ubuntu25:~/Documents/csttasks$ ./elif.sh
Enter number a
4
Enter number b
2
4 is greater than 2
ubuntuuser@ubuntu25:~/Documents/csttasks$ ./elif.sh
Enter number a
9
Enter number b
67
9 is less than 67
ubuntuuser@ubuntu25:~/Documents/csttasks$ ./elif.sh
Enter number a
6
Enter number b
6
6 is equal to 6
```

- **Logical Operator:**
  - **AND Operator (&&)**
    - Both conditions must be true for the combined condition to be true.
    - if [condition1] && [condition2]; then
      commands
      fi

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ cat and.sh
#!/bin/bash
echo "Enter Age"
read age
if [ $age -ge 18 ] && [ $age -le 60 ]
then
        echo "Eligible to work"
else
        echo "Not Eligible to work"
fi

ubuntuuser@ubuntu25:~/Documents/csttasks$ ./and.sh
Enter Age
67
Not Eligible to work
ubuntuuser@ubuntu25:~/Documents/csttasks$ ./and.sh
Enter Age
12
Not Eligible to work
ubuntuuser@ubuntu25:~/Documents/csttasks$ ./and.sh
Enter Age
45
Eligible to work
```

  - **OR Operator (| |)**
    - If any one condition is true, the combined condition is true.
      if [condition1] || [condition2]
      then
      commands
      fi

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ cat or.sh
#!/bin/bash
echo "Enter a"
read a
echo "Enter b"
read b
if (( a > 5 ||  b > 5 ))
then
        echo "Atleast one is Greater than 5"
else
        echo "Both are Less than 5"
fi
ubuntuuser@ubuntu25:~/Documents/csttasks$ ./or.sh
Enter a
3
Enter b
3
Both are Less than 5
ubuntuuser@ubuntu25:~/Documents/csttasks$ ./or.sh
Enter a
7
Enter b
4
Atleast one is Greater than 5
```

- **NOT Operator (!)**
  - Reverses the result of a condition.
  - if ! [condition]; then
    commands
    fi

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ cat not.sh
#!/bin/bash

a=3
if ! [ $a -gt 5 ]
then
        echo "Condition is false, so NOT makes it true"
fi
ubuntuuser@ubuntu25:~/Documents/csttasks$ ./not.sh
Condition is false, so NOT makes it true
```

- **Relational Operator:**
  - **-eq :** equal
    - [$a -eq $b] : true if a = b
  - **-ne :** not equal
    - [$a -ne $b] : true if a ≠ b
  - **-gt :** greater than
    - [$a -gt $b] : true if a > b

- **-lt :** less than
  - [$a -lt $b] : true if a < b
- **-ge :** greater than or equal
  - [$a -ge $b] : true if a >= b
- **-le :** less than or equal
  - [$a -le $b] → true if a <= b

- **Backup Files:**

```
ubuntuuser@ubuntu25:~/Documents/csttasks$ cat backup.sh
#!/bin/bash
timestamp=$(date +"%Y-%m-%d_%H-%M-%S")
backup_dir="backup"
mkdir -p "$backup_dir"

for file in "$@"
do
        if [ -f "$file" ]
        then
                cp "$file" "$backup_dir/${file}_$timestamp"
                echo "Backed up : $file"
        else
                echo "File Not found: $file"
        fi
done


ubuntuuser@ubuntu25:~/Documents/csttasks$ touch file10 file11 file12 file13
ubuntuuser@ubuntu25:~/Documents/csttasks$ ./backup.sh file10 file11 file12 file13
Backed up : file10
Backed up : file11
Backed up : file12
Backed up : file13
ubuntuuser@ubuntu25:~/Documents/csttasks$ ls
5     and.sh    arithm.sh  backup.sh  else.sh  file10  file12  file2  file5  file7  or.sh           sample.txt  test.sh
TASK  arith.sh  backup     elif.sh    file1    file11  file13  file3  file6  if.sh  permission.txt  task
ubuntuuser@ubuntu25:~/Documents/csttasks$ cd backup
ubuntuuser@ubuntu25:~/Documents/csttasks/backup$ ls
file10_2025-12-17_13-55-17  file11_2025-12-17_13-55-17  file12_2025-12-17_13-55-17  file13_2025-12-17_13-55-17
ubuntuuser@ubuntu25:~/Documents/csttasks/backup$
```

- "$@" : all command-line arguments passed to the script
- Each argument is assigned one by one to a variable file i.e., if run the shell script with the filenames that we want to backup (./backup.sh file1 file2 file3), every file will be taken one by one and backed up.
- If [-f “$file”] : -f is the command to check the file exists and is a regular file. If only that condition is true the file is backed up.
- ${file}_$timestamp uses curly braces for $file to separate it from _ otherwise the variable will be identified as file_.

Day 4 - 18/12/2025

Tasks Completed Today:

- **Task1:** Wrote shell scripts using FOR, WHILE, UNTIL and SELECT loops.
- **Task2:** Installed packages using apt command.
- **Task3:** Familiarized enabling/disabling universe, multiverse and restricted repository on ubuntu.
- **Task4:** Created VPC with subnets, route tables, internet gateways, nats, security group etc manually. Created Security Group and EC2 machine with t2.micro instance type and connect with the same VPC and security group created earlier.

Learnings:

- **Loops:**
  - **For loop:**
    - A for loop iterates over a fixed list or range.
    - for variable in list
      do
      commands
      done

```
ubuntuuser@ubuntu25:~/Documents/csttasks/loops$ cat for.sh
#!/bin/bash

for file in ~/Documents/csttasks/*
do
        echo "$file"
done
ubuntuuser@ubuntu25:~/Documents/csttasks/loops$ ./for.sh
/home/ubuntuuser/Documents/csttasks/5
/home/ubuntuuser/Documents/csttasks/and.sh
/home/ubuntuuser/Documents/csttasks/arithm.sh
/home/ubuntuuser/Documents/csttasks/arith.sh
/home/ubuntuuser/Documents/csttasks/backup
/home/ubuntuuser/Documents/csttasks/backup.sh
/home/ubuntuuser/Documents/csttasks/elif.sh
/home/ubuntuuser/Documents/csttasks/else.sh
/home/ubuntuuser/Documents/csttasks/file1
/home/ubuntuuser/Documents/csttasks/file10
/home/ubuntuuser/Documents/csttasks/file11
/home/ubuntuuser/Documents/csttasks/file12
/home/ubuntuuser/Documents/csttasks/file13
/home/ubuntuuser/Documents/csttasks/file2
/home/ubuntuuser/Documents/csttasks/file3
/home/ubuntuuser/Documents/csttasks/file5
/home/ubuntuuser/Documents/csttasks/file6
/home/ubuntuuser/Documents/csttasks/file7
/home/ubuntuuser/Documents/csttasks/if.sh
/home/ubuntuuser/Documents/csttasks/loops
/home/ubuntuuser/Documents/csttasks/not.sh
/home/ubuntuuser/Documents/csttasks/or.sh
/home/ubuntuuser/Documents/csttasks/permission.txt
/home/ubuntuuser/Documents/csttasks/sample.txt
/home/ubuntuuser/Documents/csttasks/task
/home/ubuntuuser/Documents/csttasks/TASK
/home/ubuntuuser/Documents/csttasks/test.sh
```

  - **While loop:**
    - A while loop runs as long as a condition is true.

- while [condition]
  do
  commands
  done

```
ubuntuuser@ubuntu25:~/Documents/csttasks/loops$ cat while.sh
#!/bin/bash
i=1
while [ $i -le 10 ]
do
        echo "$i"
        ((i++))
done

ubuntuuser@ubuntu25:~/Documents/csttasks/loops$ ./while.sh
1
2
3
4
5
6
7
8
9
10
```

- **Until loop:**
  - Opposite of while. Enters inside a loop when a condition is false and exit only when it's true.
  - until [condition]
    do
    commands
    done

```
ubuntuuser@ubuntu25:~/Documents/csttasks/loops$ cat until.sh
#!/bin/bash

until [ "$answer" = "yes" ]
do
        read -p "Type yes to continue:" answer
done


ubuntuuser@ubuntu25:~/Documents/csttasks/loops$ ./until.sh
Type yes to continue:I
Type yes to continue:AM
Type yes to continue:KAILAS
Type yes to continue:yes
ubuntuuser@ubuntu25:~/Documents/csttasks/loops$
```

- **Select loop:**
  - Show a menu and run commands based on the user's choice.
  - select variable in list
    do
      commands
    done

```
ubuntuuser@ubuntu25:~/Documents/csttasks/loops$ cat select.sh
#!/bin/bash

select option in "Check Disk" "Check Memory" "Restart Nginx" "Exit"
do
        case $option in
                "Check Disk")
                        df -h
                        ;;
                "Check Memory")
                        free -h
                        ;;
                "Restart Nginx")
                        sudo systemctl restart
                        echo "Nginx Restarted"
                        ;;
                "Exit")
                        echo "Exiting"
                        break
                        ;;
                *)
                        echo "Invalid Option"
                        ;;
                esac
        done

ubuntuuser@ubuntu25:~/Documents/csttasks/loops$ ./select.sh
1) Check Disk
2) Check Memory
3) Restart Nginx
4) Exit
#? 2
               total        used        free      shared  buff/cache   available
Mem:           3.3Gi       2.7Gi        96Mi       169Mi       853Mi       669Mi
Swap:             0B          0B          0B
#? 1
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           681M  1.7M  679M   1% /run
/dev/sda2        30G  8.4G   20G  31% /
```

- **Install Packages:**
  - apt is the command for packaging tool in ubuntu.
  - Install software, update software, remove software, manage repositories
  - Needs root privilege (sudo).
  - **apt install package_name:** Installs package. Can install more than one package if specified altogether with the command.

```
root@ubuntu25:~# apt install curl tree -y
Installing:
  curl  tree

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 15
  Download size: 309 kB
  Space needed: 622 kB / 22.5 GB available

Get:1 http://in.archive.ubuntu.com/ubuntu questing/main amd64 curl amd64 8.14.1-2ubuntu1 [259 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu questing-updates/universe amd64 tree amd64 2.2.1-1ubuntu0.2 [49.9 kB]
Fetched 309 kB in 2s (180 kB/s)
Selecting previously unselected package curl.
(Reading database ... 145888 files and directories currently installed.)
Preparing to unpack .../curl_8.14.1-2ubuntu1_amd64.deb ...
Unpacking curl (8.14.1-2ubuntu1) ...
Selecting previously unselected package tree.
Preparing to unpack .../tree_2.2.1-1ubuntu0.2_amd64.deb ...
Unpacking tree (2.2.1-1ubuntu0.2) ...
Setting up tree (2.2.1-1ubuntu0.2) ...
Setting up curl (8.14.1-2ubuntu1) ...
Processing triggers for man-db (2.13.1-1) ...
```

- **apt update:** Updates package list
- **apt upgrade:** Updates installed packages to newer versions.
- **apt remove package_name:** Removes a package

```
root@ubuntu25:~# apt remove curl
REMOVING:
  curl

Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 15
  Freed space: 505 kB

Continue? [Y/n] y
(Reading database ... 145904 files and directories currently installed.)
Removing curl (8.14.1-2ubuntu1) ...
Processing triggers for man-db (2.13.1-1) ...
```

- **apt purge package_name:** Removes package along with its config files.
- **apt search package_name:** Searches for a package.
- **apt show package_name:** Shows details regarding a package.

```
Setting up curl (8.14.1-2ubuntu1) ...
Processing triggers for man-db (2.13.1-1) ...
root@ubuntu25:~# apt show tree
Package: tree
Version: 2.2.1-1ubuntu0.2
Priority: optional
Section: universe/utils
Origin: Ubuntu
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Original-Maintainer: Florian Ernst <florian@debian.org>
Bugs: https://bugs.launchpad.net/ubuntu/+filebug
Installed-Size: 117 kB
Depends: libc6 (>= 2.38)
Homepage: http://oldmanprogrammer.net/source.php?dir=projects/tree
Task: xubuntu-desktop, lubuntu-desktop, ubuntu-mate-core, ubuntu-mate-desktop, ubuntu-budgie-desktop-minimal, ubuntu-budgie-desktop, ubuntu-budgie-d
Download-Size: 49.9 kB
APT-Manual-Installed: yes
APT-Sources: http://in.archive.ubuntu.com/ubuntu questing-updates/universe amd64 Packages
Description: displays an indented directory tree, in color
 Tree is a recursive directory listing command that produces a depth indented
 listing of files, which is colorized ala dircolors if the LS_COLORS environment
 variable is set and output is to tty.

Notice: There is 1 additional record. Please use the '-a' switch to see it
```

- **Enabling/Disabling Repositories:**
  - Two methods of enabling/disabling repositories. Either using **apt-add-repository** or /etc/apt/sources.list.d/ubuntu.sources can be edited. This file contains the list of enabled repositories. If the name of

the specific repository is removed from the file, the installation of packages belonging to that repository will be restricted.

- **apt-add-repository --remove repository_name:** This will disable the repository. There won't be any output if we grep and look for the repository type (main, restricted, universe, multiverse).

```
Types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: questing-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

root@ubuntu25:~# apt-add-repository --remove restricted
Removing component(s) 'restricted' from all repositories.
Press [ENTER] to continue or Ctrl-c to cancel.
Removed restricted from: <child deb http://in.archive.ubuntu.com/ubuntu/ questing of <weakref at 0x7c707706bb50; to 'aptsources.sourc
ceEntry' at 0x7c707729f100>
Removed restricted from: <child deb http://in.archive.ubuntu.com/ubuntu/ questing-updates of <weakref at 0x7c707706bc40; to 'aptsourc
b822SourceEntry' at 0x7c707729eea0>
Removed restricted from: <child deb http://in.archive.ubuntu.com/ubuntu/ questing-backports of <weakref at 0x7c707706bd30; to 'aptsou
Deb822SourceEntry' at 0x7c7077115bd0>
Removed restricted from: Types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: questing-security
Components: universe main multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
```

```
root@ubuntu25:~# apt-add-repository --list | grep restricted
root@ubuntu25:~# apt update
Hit:1 http://in.archive.ubuntu.com/ubuntu questing InRelease
Hit:2 http://security.ubuntu.com/ubuntu questing-security InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu questing-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu questing-backports InRelease
15 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ubuntu25:~# apt install nvidia-driver-525
Error: Unable to locate package nvidia-driver-525
root@ubuntu25:~#
```

- **apt-add-repository repository_name:** This will enable the repository type. (restricted, universe, multiverse).

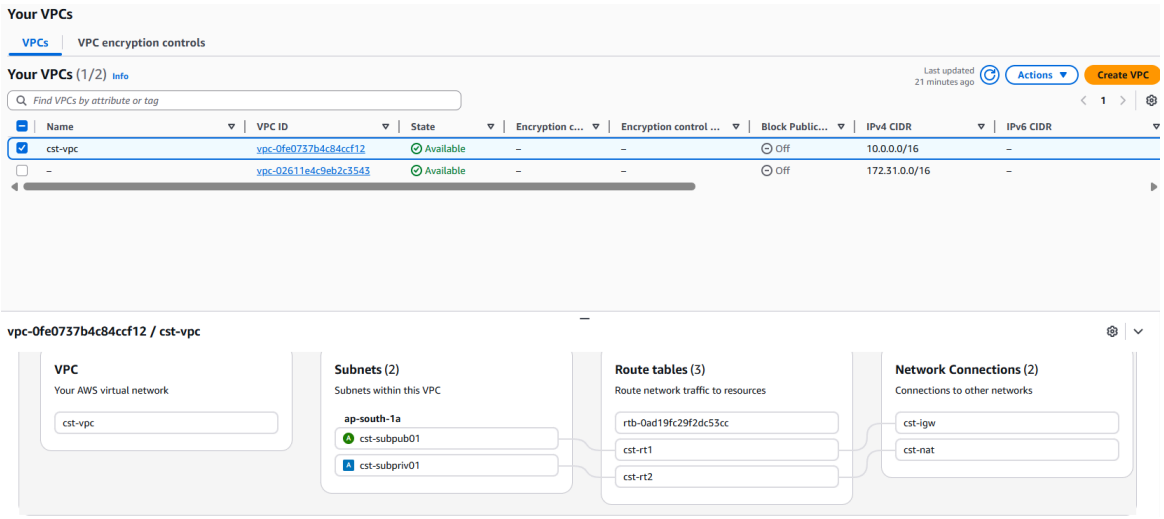
```
root@ubuntu25:~# apt-add-repository restricted
Adding component(s) 'restricted' to all repositories.
Press [ENTER] to continue or Ctrl-c to cancel.
Added restricted to: <child deb http://in.archive.ubuntu.com/ubuntu/ questing of <weakref at 0x7dc05745f9c0; to 'aptsources.sourceslis
try' at 0x7dc057693100>
Added restricted to: Types: deb
URIs: http://in.archive.ubuntu.com/ubuntu/
Suites: questing-updates
Components: universe main multiverse restricted
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
Added restricted to: Types: deb
URIs: http://in.archive.ubuntu.com/ubuntu/
Suites: questing-backports
Components: universe main multiverse restricted
```

```
root@ubuntu25:~# apt-add-repository --list | grep restricted
Components: universe main multiverse restricted
Components: universe main multiverse restricted
root@ubuntu25:~#
```

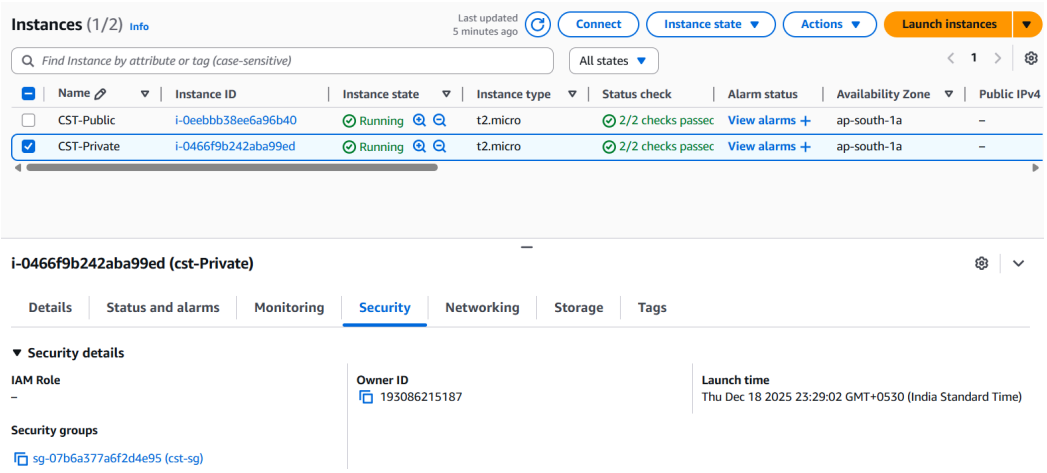
```
root@ubuntu25:~# apt install nvidia-driver-575 -y
Installing:
  nvidia-driver-575

Installing dependencies:
  build-essential          libalgorithm-diff-perl     libnvidia-common-580    libvdpau1                    nvidia-utils-580
  bzip2                    libalgorithm-diff-xs-perl  libnvidia-compute-580   libxnvctrl0                  ocl-icd-libopencl1
  dkms                     libalgorithm-merge-perl    libnvidia-decode-580    lto-disabled-list            pkg-config
  dpkg-dev                 libasan8                   libnvidia-egl-wayland1  make                         pkgconf
  fakeroot                 libcc1-0                   libnvidia-encode-580    mesa-vdpau-drivers           pkgconf-bin
  g++                      libdpkg-perl               libnvidia-extra-580     nvidia-compute-utils-580     screen-resolution-extra
  g++-15                   libfakeroot                libnvidia-fbc1-580      nvidia-dkms-580              vdpau-driver-all
  g++-15-x86-64-linux-gnu  libfile-fcntllock-perl     libnvidia-gl-580        nvidia-driver-580            xcvt
  g++-x86-64-linux-gnu     libgcc-15-dev              libpkgconf3             nvidia-firmware-580-580.95.05  xserver-xorg-core
  gcc                      libhwasan0                 libquadmath0            nvidia-kernel-common-580     xserver-xorg-video-nvidia-580
  gcc-15                   libitm1                    libstdc++-15-dev        nvidia-kernel-source-580
  gcc-15-x86-64-linux-gnu  liblsan0                   libtsan2                nvidia-prime
  gcc-x86-64-linux-gnu     libnvidia-cfg1-580         libubsan1               nvidia-settings

Suggested packages:
  bzip2-doc       debian-tag2upload-keyring  gcc-15-doc    automake  bison            gcc-15-locales       libstdc++-15-doc  xfonts-100dpi
  menu            g++-multilib               gcc-multilib  libtool   gcc-doc          gdb-x86-64-linux-gnu  make-doc          | xfonts-75dpi
  debian-keyring  g++-15-multilib            autoconf      flex      gcc-15-multilib  bzr                  libvdpau-va-gl1   xfonts-scalable

Recommended packages:
  libnvidia-compute-580:i386  libnvidia-decode-580:i386  libnvidia-encode-580:i386  libnvidia-fbc1-580:i386  libnvidia-gl-580:i386

Summary:
  Upgrading: 0, Installing: 63, Removing: 0, Not Upgrading: 15
  Download size: 459 MB
  Space needed: 1,358 MB / 22.5 GB available
```

- After we disable any repository we won't be able to install a package belonging to that repository unless we enable it. After enable/disable its mandatory to update the package list using apt update.

- **VPC and EC2 Instance Creation:**
  - Created VPC. Name: cst-vpc
  - Created subnets for both public and private access. Name: cst-subpub01 (public) cst-subpriv01 (private).
  - Create Internet Gateway (IGW) for the public subnet. Attached to VPC created earlier.
  - Created Route Table (cst-rt1). Added route and set destination to IGW. Associated the Public Subnet that was created earlier.
  - Allocated Elastic IP so that private subnet instances can access the internet using a fixed, public IPv4 address.
  - Created NAT gateway for the private subnet. Set Connectivity type to Public. Assigned the Public Subnet for external access. Allocated the Elastic IP.
  - Created Route Table (cst-rt2) for private subnet. Added routes to set destination to NAT gateway. Associated private subnet created earlier.



- Created security group. Added SSH and HTTP as inbound rule and All Traffic as outbound rule.
- Created an EC2 instance and allocated the public subnet in the VPC created earlier. Allocated the security group created earlier.



- Created a Private EC2 instance. Used the Public EC2 as the bastion host. Copied the key pair into the Public EC2 and connected to private EC2 user@private_ip from the public EC2 using ssh.

```
C:\Users\kaila\Downloads>ssh -i "CST.pem" ec2-user@10.0.2.212
ssh: connect to host 10.0.2.212 port 22: Connection timed out

C:\Users\kaila\Downloads>ssh -i "CST.pem" ec2-user@13.201.27.181
The authenticity of host '13.201.27.181 (13.201.27.181)' can't be established.
ED25519 key fingerprint is SHA256:hXqhy8UH+4o1a/cmwjvskiInaiYyx9i62lgNbimFGpM.
This host key is known by the following other names/addresses:
    C:\Users\kaila/.ssh/known_hosts:38: 3.110.212.243
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.201.27.181' (ED25519) to the list of known hosts.
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Thu Dec 18 12:26:24 2025 from 103.177.253.70
[ec2-user@ip-10-0-1-40 ~]$ ls
[ec2-user@ip-10-0-1-40 ~]$ vim CST.pem
[ec2-user@ip-10-0-1-40 ~]$ chmod 400 CST.pem
[ec2-user@ip-10-0-1-40 ~]$ ssh -i "CST.pem" ec2-user@10.0.2.212
The authenticity of host '10.0.2.212 (10.0.2.212)' can't be established.
ED25519 key fingerprint is SHA256:G8WPRs/1oTessW/S6JSl6a2jQ2s4FbpBdZjv6U6PUdM.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.212' (ED25519) to the list of known hosts.
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-2-212 ~]$ |
```